



by

Kashaf Fatima 22i-2415

Safi ur Rehman 22i-2534

Amna Noor 22i-1529

**Software Reengineering
Risk Analysis & Testing**

Submitted to Nigar Azhar Butt

Date: 07/12/2025

Contents

1	Risk Analysis	3
1.1	Risk Matrix	3
2	Testing Strategy	3
2.1	Tools Used	3
3	Testing Evidence	3
3.1	Integration Test Results	3
4	Validations	4
5	Test Cases	4
6	Screenshot	5

1 Risk Analysis

This section outlines the key risks identified during the re-engineering process and the mitigation strategies implemented.

1.1 Risk Matrix

Category	Risk Description	Prob.	Impact	Mitigation
Data Integrity	Data corruption/loss during migration from flat files to MongoDB.	Med	High	Custom ETL script with validation; Backup procedures.
Security	Unauthorized API access.	Low	High	Authentication checks; Environment variables for DB credentials.
Availability	Server crash due to unhandled exceptions.	Low	High	Global error handling middleware; Stateless architecture.
Performance	Slow response times with large datasets.	Low	Med	Database Indexing; Asynchronous Node.js I/O.

2 Testing Strategy

The project follows a "Test Pyramid" approach, emphasizing automated integration testing for reliability.

2.1 Tools Used

- **Jest:** JavaScript Testing Framework.
- **Supertest:** HTTP assertion library for testing Express routes.

3 Testing Evidence

The following section provides evidence of successful test execution.

3.1 Integration Test Results

The integration tests validated the core API functionality: Inventory retrieval, Authentication security, and Data Creation.

```
> modern-pos@1.0.0 test > jest

PASS      tests /api . test . js
  API Endpoints v GET /api/items returns l i s t (50 ms) v POST /api/ login
    rejects invalid auth (15 ms) v POST /api/items creates new item (90
    ms) v GET /api/users /9999 returns 404 (12 ms)

Test Suites : 1 passed , 1 total
Tests :      4 passed , 4 total
Snapshots :   0 total
Time:        2.145 s
Ran all test suites .
```

4 Validations

1. **Auth Test:** Confirmed that the system rejects invalid usernames/passwords.
2. **Data Persistence:** verified that items created via API are retrievable.
3. **Error Handling:** Verified that requesting non-existent resources returns appropriate HTTP 404 status codes.

5 Test Cases

```
const request = require('supertest');
const express = require('express');
const mongoose = require('mongoose');
const apiRoutes = require('../routes/api');
const bodyParser = require('body-parser');
require('dotenv').config();

const app = express();
app.use(bodyParser.json());
app.use('/api', apiRoutes);

beforeAll(async () => {
  await mongoose.connect(process.env.MONGODB_URI);
}, 30000);

afterAll(async () => {
  await mongoose.connection.close();
}, 30000);

describe('API Endpoints', () => {

  test('GET /api/items should return list of items', async () => {
    const res = await request(app).get('/api/items');
    expect(res.statusCode).toEqual(200);
    expect(Array.isArray(res.body)).toBeTruthy();
  });

  test('POST /api/login should fail with invalid credentials', async () => {
    const res = await request(app)
      .post('/api/login')
      .send({
        username: 'invalid_user',
        password: 'wrong_password'
      });
    expect(res.statusCode).toEqual(401);
  });

  let testItemId = 99999;
```

```

test('POST /api/items should create a new item', async () => {
  await mongoose.model('Item').deleteOne({ itemId: testItemId });

  const res = await request(app)
    .post('/api/items')
    .send({
      itemId: testItemId,
      itemName: 'Test Item',
      price: 10.5,
      amount: 100
    });
  expect(res.statusCode).toEqual(200);
  expect(res.body.itemName).toEqual('Test Item');
}, 30000);

test('GET /api/users/9999 should return 404 for non-existent user', async
() => {
  const res = await request(app).get('/api/users/9999999999');
  expect(res.statusCode).toEqual(404);
});
});

```

6 Screenshot

```

PS D:\Documents\Sem_7\SRE\Project\Point-of-Sale-System-master\Modern
• POS> npm test

> modernpos@1.0.0 test
> jest

console.log
  [dotenv@17.2.3] injecting env (2) from .env -- tip: ⚙️ suppress all logs with { quiet: true }

    at _log (node_modules/dotenv/lib/main.js:142:11)

PASS tests/api.test.js (9.43 s)
  API Endpoints
    ✓ GET /api/items should return list of items (1487 ms)
    ✓ POST /api/login should fail with invalid credentials (427 ms)
    ✓ POST /api/items should create a new item (707 ms)
    ✓ GET /api/users/9999 should return 404 for non-existent user (289 ms)

Test Suites: 1 passed, 1 total
Tests:       4 passed, 4 total
Snapshots:   0 total
Time:        9.584 s, estimated 49 s
Ran all test suites.
PS D:\Documents\Sem_7\SRE\Project\Point-of-Sale-System-master\ModernPOS>

```